

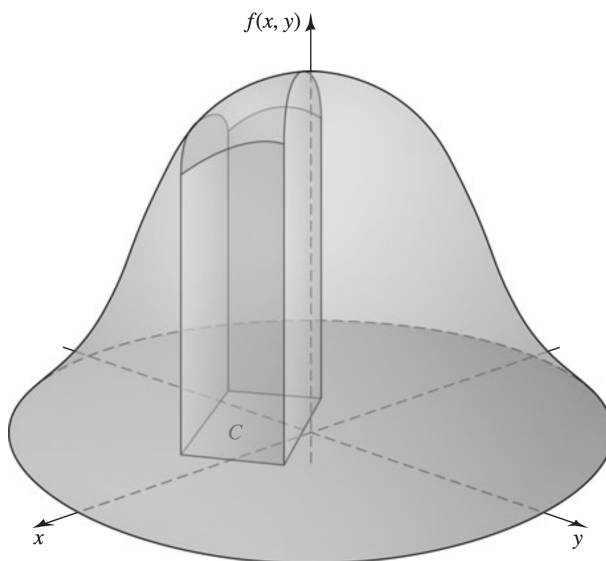


Figure 2.2: A bivariate density is a surface in three dimensional space, and probabilities are volumes beneath the surface. The total volume beneath the surface must be one.

## 2.2 Jointly absolutely continuous random pairs

**Definition 2.6.** If  $(X, Y)$  are jointly absolutely continuous, then their joint distribution can be summarized with a **joint probability density function (joint pdf)**. This is a bivariate function  $f_{XY} : \mathbb{R}^2 \rightarrow \mathbb{R}$  with the following properties:

- $f_{XY}(x, y) \geq 0$ ;
- $P((X, Y) \in \mathbb{R}^2) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{XY}(x, y) dx dy = 1$ .

Given the joint pdf, you can compute the joint distribution by integrating:

$$P((X, Y) \in C) = \int_C \int f_{XY}(x, y) dx dy, \quad C \subseteq \mathbb{R}^2.$$

In the univariate case, probability corresponds to area under the density curve. In the bivariate case, probability corresponds to volume under the density *surface*, as in Figure 2.2.

**Definition 2.7.** If  $(X, Y)$  are jointly absolutely continuous, then each random variable  $X$  and  $Y$  has a **marginal probability density function (marginal pdf)** that describes the distribution

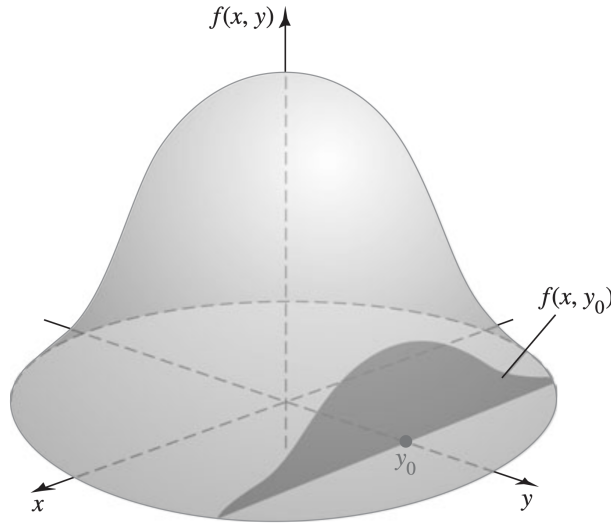


Figure 2.3: The conditional density is a renormalized cross-section of the joint density, where we hold the argument of the conditioning variable constant.

of that random variable on its own:

$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x, y) dy \quad (2.13)$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x, y) dx. \quad (2.14)$$

To compute  $f_X$ , for example, we say that we *integrate y out* of the joint density.

To see where the marginal density formulas come from, consider the marginal CDF of one of the variables:

$$F_X(x) = P(X \leq x) = \int_{-\infty}^x \int_{-\infty}^{\infty} f_{XY}(t, y) dy dt.$$

The marginal PDF is just the derivative, so the fundamental theorem of calculus tells us that

$$f_X(x) = F'_X(x) = \frac{d}{dx} \int_{-\infty}^x \int_{-\infty}^{\infty} f_{XY}(t, y) dy dt = \int_{-\infty}^{\infty} f_{XY}(x, y) dy.$$

**Definition 2.8.** The **conditional probability density function (conditional pdf)** of  $X$  or  $Y$  describes its revised behavior *conditional* on the event  $Y = y$  or  $X = x$ , respectively. They are defined to be

$$f_{X|Y}(x | y) = \frac{f_{XY}(x, y)}{f_Y(y)} \quad (2.15)$$

$$f_{Y|X}(y | x) = \frac{f_{XY}(x, y)}{f_X(x)}. \quad (2.16)$$

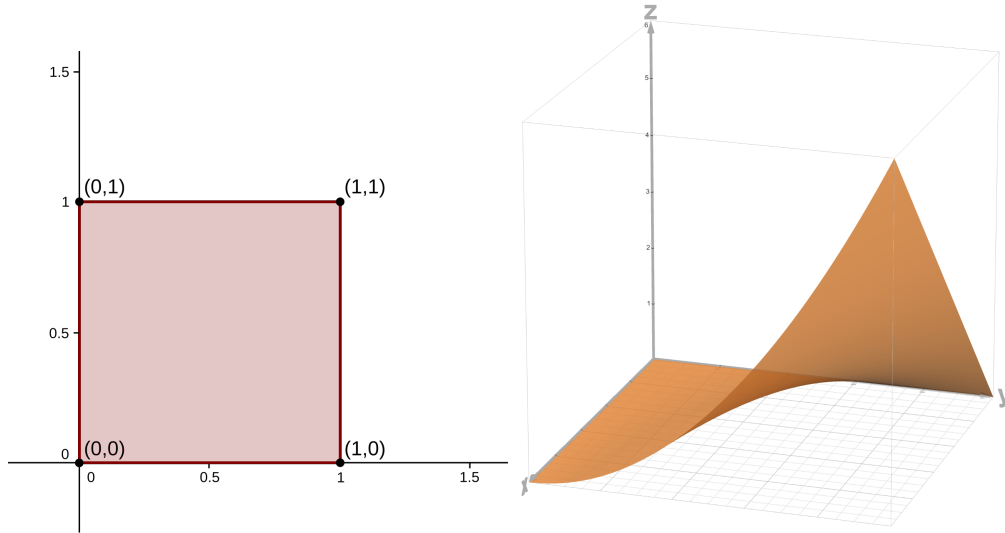


Figure 2.4: In Example 2.4, the joint range is the unit square on the left, and the joint density is the surface on the right.

So  $f_{X|Y}(x | y)$ , for example, is the cross-section of  $f_{XY}$  that holds the second argument fixed at  $y$  but lets  $x$  vary. The denominator renormalizes to ensure that the new *conditional* density integrates to 1 in  $x$ . See Figure 2.3 for a cartoon.

**Theorem 2.4. (Marginal-conditional decomposition)** From Definition 2.8, we get “two identities for the price of one:”

$$f_{XY}(x, y) = f_{X|Y}(x | y)f_Y(y) = f_{Y|X}(y | x)f_X(x). \quad (2.17)$$

**Corollary 2.5. (Hierarchical representation)** In order to specify a joint distribution for  $(X, Y)$ , you can do so *hierarchically*:

$$\begin{aligned} X &\sim P_X \\ Y | X &\sim P_{Y|X}. \end{aligned}$$

Theorem 2.4 guarantees that we can stitch these two pieces together to get a valid joint distribution.

**Theorem 2.6. (Bayes’ theorem, again)** Combining the definition of conditional density with the marginal-conditional decomposition gives a version of Bayes’ theorem for the conditional PDFs:

$$f_{X|Y}(x | y) = \frac{f_{XY}(x, y)}{f_Y(y)} = \frac{f_{Y|X}(y | x)f_X(x)}{f_Y(y)} \quad (2.18)$$

$$f_{Y|X}(y | x) = \frac{f_{XY}(x, y)}{f_X(x)} = \frac{f_{X|Y}(x | y)f_Y(y)}{f_X(x)}. \quad (2.19)$$

**Example 2.4.** Let  $X$  and  $Y$  have joint density

$$f_{XY}(x, y) = \begin{cases} 6xy^2, & x, y \in (0, 1) \\ 0 & \text{else.} \end{cases}$$

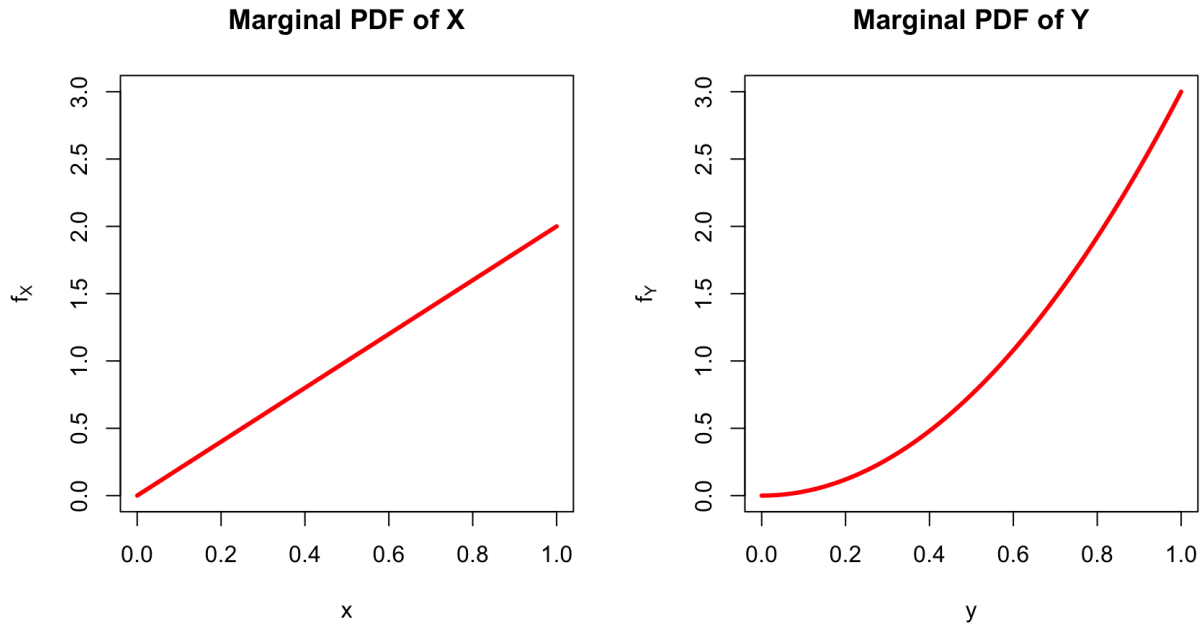


Figure 2.5: Marginal densities in Example 2.4

The joint range is  $\text{Range}(X, Y) = \text{supp}(f_{XY}) = (0, 1) \times (0, 1)$ , the unit square in the plane (Figure 2.4). The marginal densities are

$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x, y) dy = \int_0^1 6xy^2 dy = [2xy^3]_0^1 = 2x, \quad x \in (0, 1).$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x, y) dx = \int_0^1 6xy^2 dx = [3x^2y^2]_0^1 = 3y^2, \quad y \in (0, 1).$$

The conditional densities are

$$f_{X|Y}(x|y) = \frac{f_{XY}(x, y)}{f_Y(y)} = \frac{6xy^2}{3y^2} = 2x, \quad x \in (0, 1)$$

$$f_{Y|X}(y|x) = \frac{f_{XY}(x, y)}{f_X(x)} = \frac{6xy^2}{2x} = 3y^2, \quad y \in (0, 1).$$

We see that  $f_{X|Y} = f_X$  and  $f_{Y|X} = f_Y$ , meaning that conditioning does not actually teach us anything. This motivates a definition.

**Definition 2.9.** A jointly absolutely continuous pair  $(X, Y)$  are **independent** if

$$f_{XY}(x, y) = f_X(x)f_Y(y) \quad \text{for all } (x, y) \in \mathbb{R}^2. \quad (2.20)$$

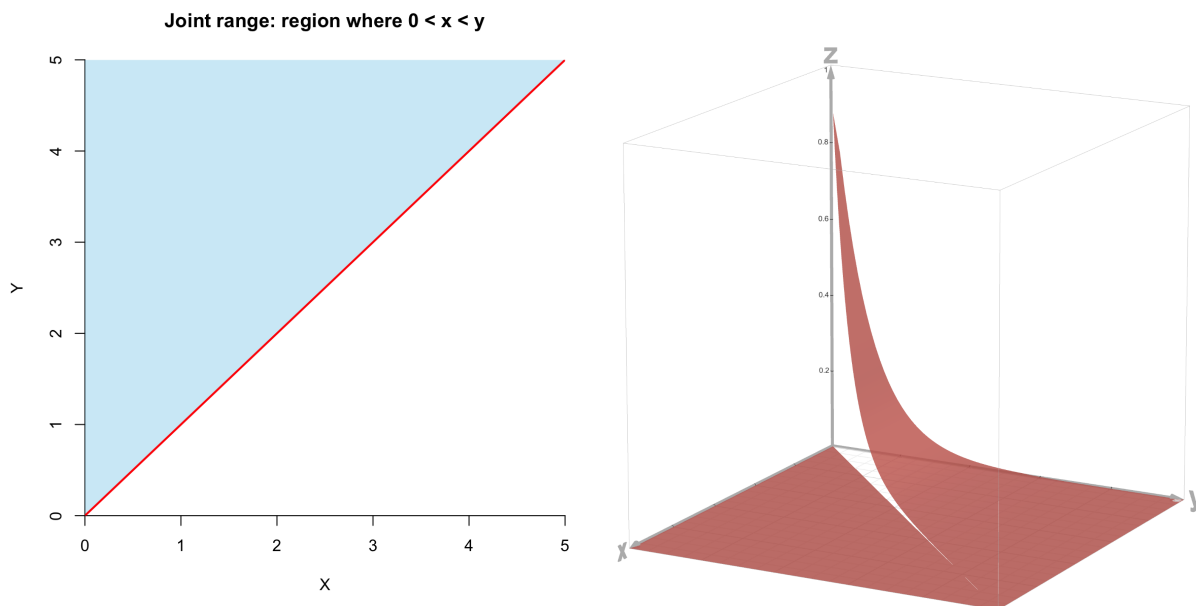


Figure 2.6: The joint density in Example 2.5.

Alternatively but equivalently,  $X$  and  $Y$  are independent if

$$f_{X|Y}(x | y) = f_X(x) \quad \text{for all } (x, y) \in \mathbb{R}^2 \quad (2.21)$$

$$f_{Y|X}(y | x) = f_Y(y) \quad \text{for all } (x, y) \in \mathbb{R}^2. \quad (2.22)$$

**Example 2.5.** Consider  $(X, Y)$  with joint density

$$f_{XY}(x, y) = \begin{cases} e^{-y} & 0 < x < y < \infty \\ 0 & \text{else.} \end{cases}$$

The joint range and the density surface are displayed in Figure 2.6. Let's compute all of the marginals and conditionals:

$$\begin{aligned} f_X(x) &= \int_{-\infty}^{\infty} f_{XY}(x, y) dy = \int_x^{\infty} e^{-y} dy = [-e^{-y}]_x^{\infty} = e^{-x}, & 0 < x \\ f_{Y|X}(y | x) &= \frac{f_{XY}(x, y)}{f_X(x)} = \frac{e^{-y}}{e^{-x}} = e^{-(y-x)}, & x < y \\ f_Y(y) &= \int_{-\infty}^{\infty} f_{XY}(x, y) dx = \int_0^y e^{-y} dx = e^{-y} \int_0^y dx = ye^{-y}, & 0 < y \\ f_{X|Y}(x | y) &= \frac{f_{XY}(x, y)}{f_Y(y)} = \frac{e^{-y}}{ye^{-y}} = \frac{1}{y}, & 0 < x < y. \end{aligned}$$

These are all familiar distributions:

$$X \sim \text{Exponential}(1)$$

$$Y \sim \text{Gamma}(2, 1)$$

$$X | Y = y \sim \text{Unif}(0, y).$$

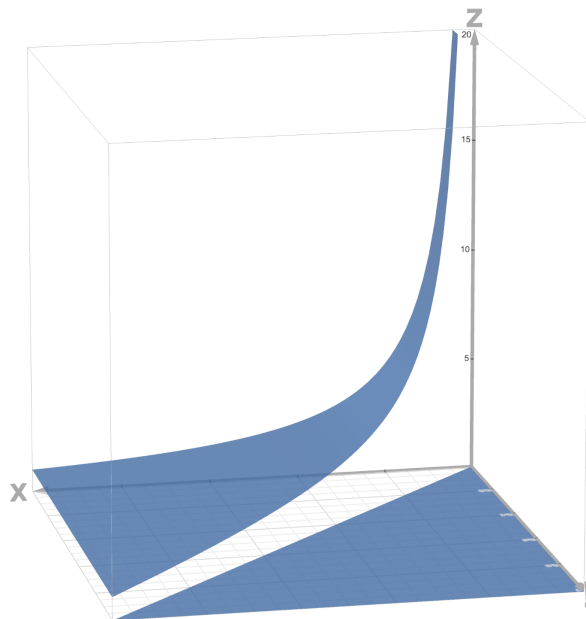


Figure 2.7: Joint density in Example 2.6.

Conditional on  $X = x$ ,  $Y$  is Exponential(1), but shifted right by the amount  $x$ .

**Example 2.6.** Consider a joint distribution for  $(X, Y)$  written hierarchically:

$$\begin{aligned} X &\sim \text{Unif}(0, 1) \\ Y \mid X = x &\sim \text{Unif}(0, x). \end{aligned}$$

So

$$\begin{aligned} f_X(x) &= 1 & 0 \leq x \leq 1 \\ f_{Y|X}(y \mid x) &= \frac{1}{x} & 0 \leq y \leq x \\ f_{XY}(x, y) &= f_{Y|X}(y \mid x)f_X(x) \\ &= \frac{1}{x} & 0 \leq y \leq x \leq 1. \end{aligned}$$

The joint density surface is displayed in Figure 2.7. The marginal density of  $Y$  is

$$\begin{aligned}
 f_Y(y) &= \int_0^1 f_{XY}(x, y) dx \\
 &= \int_0^y f_{XY}(x, y) dx + \int_y^1 f_{XY}(x, y) dx \\
 &= \int_0^y 0 dx + \int_y^1 \frac{1}{x} dx \\
 &= \int_y^1 \frac{1}{x} dx \\
 &= [\ln x]_y^1 \\
 &= \ln 1 - \ln y \\
 &= -\ln y, \qquad 0 < y < 1.
 \end{aligned}$$

The conditional density is

$$f_{X|Y}(x | y) = \frac{f_{XY}(x, y)}{f_Y(y)} = \frac{1/x}{-\ln y}, \quad 0 \leq y \leq x \leq 1.$$

To compute the conditional expectation of  $X$  given  $Y = y$ , we apply the definition of expected value, but we use the *conditional* density:

$$\begin{aligned}
 E(X | Y = y) &= \int_0^1 x f_{X|Y}(x | y) dx \\
 &= \int_y^1 x f_{X|Y}(x | y) dx \\
 &= \int_y^1 x \frac{1/x}{-\ln y} dx \\
 &= \frac{1}{-\ln y} \int_y^1 dx \\
 &= \frac{1 - y}{-\ln y}.
 \end{aligned}$$

**Example 2.7.** Consider the joint distribution of random variables  $X$  and  $Y$ , written in hierarchical form:

$$\begin{aligned} X &\sim \text{Gamma}\left(\frac{\nu}{2}, \frac{\nu}{2}\right) \\ Y | X = x &\sim N(0, 1/x). \end{aligned}$$

This models a normal variable with a “random variance”. To derive the marginal pdf of  $Y$ , we fix  $y \in \mathbb{R}$ , and get ready for some hardcore “massage and squint”:

$$\begin{aligned} f_Y(y) &= \int_0^\infty f_{Y|X}(y|x) f_X(x) dx = \int_0^\infty \underbrace{\frac{1}{\sqrt{2\pi(1/x)}} e^{-\frac{1}{2} \frac{y^2}{1/x}}}_{\text{pdf of } N(0, 1/x)} \underbrace{\frac{(\nu/2)^{\nu/2}}{\Gamma(\nu/2)} x^{\frac{\nu}{2}-1} e^{-\frac{\nu}{2}x}}_{\text{pdf of } \text{Gamma}(\nu/2, \nu/2)} dx \\ &= \frac{(\nu/2)^{\nu/2}}{\sqrt{2\pi} \Gamma(\nu/2)} \int_0^\infty x^{\frac{\nu+1}{2}-1} e^{-\frac{1}{2}(\nu+y^2)x} dx = \frac{(\nu/2)^{\nu/2}}{\sqrt{2\pi} \Gamma(\nu/2)} \frac{\Gamma((\nu+1)/2)}{[(\nu+y^2)/2]^{\frac{\nu+1}{2}}} \\ &= \frac{(\nu/2)^{\nu/2} \Gamma((\nu+1)/2)}{\sqrt{2\pi} \Gamma(\nu/2) (\nu/2)^{\frac{\nu+1}{2}}} \left(1 + \frac{1}{\nu} y^2\right)^{-\frac{\nu+1}{2}} = \frac{\Gamma((\nu+1)/2)}{\sqrt{\nu\pi} \cdot \Gamma(\nu/2)} \left(1 + \frac{1}{\nu} y^2\right)^{-\frac{\nu+1}{2}}. \end{aligned}$$

In the first equality, we integrated only over  $x \geq 0$  because that is where the pdf  $f_X(x)$  is nonzero. To get the second equality in the second line, we used the identity  $\int_0^\infty x^{a-1} e^{-bx} dx = \Gamma(a)/b^a$  which we know by the fact that the  $\text{Gamma}(a, b)$  density  $f(x) = \frac{b^a}{\Gamma(a)} x^{a-1} e^{-bx}$  integrates to one.

**Definition 2.10.** An absolutely continuous random variable  $X$  has **Student’s  $t$  distribution** on  $\text{Range}(X) = \mathbb{R}$  if its pdf is

$$\frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi} \cdot \Gamma\left(\frac{\nu}{2}\right)} \left(1 + \frac{1}{\nu} x^2\right)^{-\frac{\nu+1}{2}}, \quad x \in \mathbb{R}. \quad (2.23)$$

The parameter  $\nu > 0$  is called the **degrees of freedom**. We denote this  $X \sim t_\nu$ .

It turns out that the Cauchy distribution is the special case of Student’s  $t$  when  $\nu = 1$ :

$$f_X(x) = \frac{\Gamma(1)}{\Gamma(1/2)\sqrt{\pi}} (1+x^2)^{-1} = \frac{1}{\pi} \frac{1}{(1+x^2)}.$$

This follows because  $\Gamma(1) = 1$  and  $\Gamma(1/2) = \sqrt{\pi}$ . We see in Figure 2.8 that members of this family have a bell-shaped pdf with heavier tails than the normal, but the tails get thinner as  $\nu$  increases. In fact, as  $\nu \rightarrow \infty$ ,  $t_\nu$  converges to  $N(0, 1)$ . Table 2.3 describes the situation with the moments. We know that for  $\nu = 1$ , the Cauchy distribution has no finite moments. As  $\nu$  increases and the tails become less heavy, we accumulate more and more finite moments. In the extreme case of  $t_\infty = N(0, 1)$ , finally every moment is finite. But for finite  $0 < \nu < \infty$ , we eventually run out. So the mgf of Student’s  $t$  distribution is undefined.



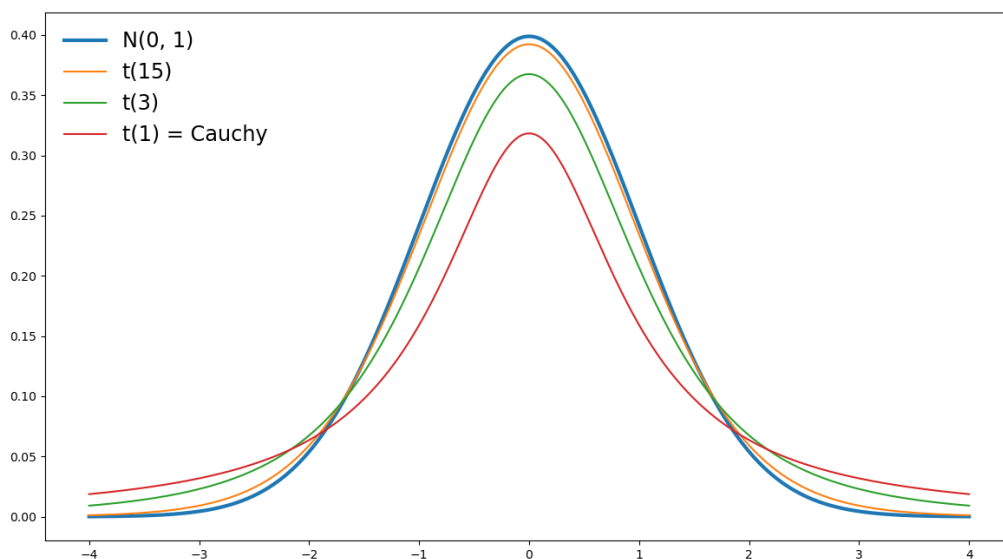


Figure 2.8

| feature          | when does it exist? | value when it exists |
|------------------|---------------------|----------------------|
| $E(X)$           | $\nu > 1$           | 0                    |
| $\text{var}(X)$  | $\nu > 2$           | $\frac{\nu}{\nu-2}$  |
| $\text{skew}(X)$ | $\nu > 3$           | 0                    |
| $\vdots$         | $\vdots$            | $\vdots$             |

Table 2.3: As the degrees of freedom of Student's  $t$  increase, the tails become thinner, and the higher moments become finite.



Figure 2.9: William Gosset (1876 - 1937) published on the  $t$  distribution under the pseudonym *Student* while he was an employee of the Guinness brewery in Ireland.